

What is Swarm? tl;dr

UNSTOPPABLE STORAGE

Swarm is a **decentralised storage and communication network** run by peer-to-peer **Bee nodes**. Upload data, get back a **content hash** — a permanent address derived from the bytes themselves. Anyone with the hash can retrieve the content from any Bee node or public gateway.

Storage is bought as **postage stamps**: prepaid batches purchased with xBZZ on Gnosis Chain. Batches have a TTL and expire unless topped up. If every batch covering a chunk expires, that content can be pruned.

HOMEPAGE Swarm  ethswarm.org	DOCS Documentation  docs.ethswarm.org	EXPLORER SwarmScan  swarmscan.io
--	---	--

What you can build on Swarm

BUILD WITH FREEDOM · SCALE WITHOUT LIMITS

File & media storage

Fixed, content-addressed URLs for any file — images, video, PDFs, datasets, ML artifacts, NFT assets.

Live video & streaming

Early proof-of-concept work on HLS over Swarm exists — expect to integrate and tune it yourself.

Decentralised websites

Publish a static front-end straight to the network. No server, no CDN, no DNS lock-in.

Chat, comments & messaging

Real-time chat, threaded comments and peer messaging. Real-time delivery needs a GSOC subscription (see back).

Blogs & dashboards

Stable URLs whose contents update via feeds. Personal blogs, public dashboards, status pages.

dApp front-ends

Host a static front-end for an Ethereum dApp on Swarm. On-chain contracts with content-addressed UI for a fully decentralised stack.

What Swarm is *not*

- **Not a database.** Opaque blobs addressed by hash — no queries, no indexing, no search. Model structure in the app layer.
- **Not permanent without payment.** Content lives only while a valid postage stamp covers its chunks.
- **Not instantly consistent.** Uploads propagate asynchronously. `deferred:true` (default) returns once data is pushed; `deferred:false` waits for read-back.
- **Not free to write.** Every upload needs a stamp funded with xBZZ. Reads have a small free tier on gateways.
- **Not a smart-contract platform.** It stores data, it does not execute code. Complements Ethereum, does not replace it.

Limitations & gotchas

- **Stamp batch expiry.** Past expiry, chunks become eligible for pruning — not deleted instantly, but can vanish. Monitor and top up.
- **Dynamic pricing.** Stamp cost tracks a price that can rise — top up so batches outlast your needs.
- **Ultra-light nodes can't upload.** Writing needs at least a light node with funds.
- **Feed writes are append-only.** Each index slot writes once; always resolve the current index before writing.
- **Capacity & lifetime are non-deterministic.** Both depend on chunking and dynamic pricing — monitor and top up.

QUICKSTART FOR SWARM DEVELOPMENT

The recommended way to start · ~5 min

1 Install Claude Code

MACOS / LINUX / WSL
`curl -fsSL https://claude.ai/install.sh | bash`
 WINDOWS · POWERSHELL
`irm https://claude.ai/install.ps1 | iex`
 Docs: code.claude.com/docs

2 Add the Swarm skills

`claude` # then sign in via browser (OAuth)
`git clone https://github.com/ethersphere/swarm-quickstart-skills.git`
`cd swarm-quickstart-skills/.claude`
`claude`

3 Run `/swarm` to set up

Runs a guided setup — checks for an existing node and installs/configures Bee if needed.

SUGGESTED DEVELOPER JOURNEY

`/swarm` → `/swarm-setup-bee-interactive` → `/swarm-stamps` → `/swarm-upload-download` | `/swarm-host-website` | `/swarm-blog`

Type `/swarm` anytime for the full menu.



Prefer written docs? → justdeploy.bzz.link

Step-by-step guides for getting started on Swarm — Bee node install, funding, site hosting — with beginner, intermediate and advanced tracks. Mirror: justdeploy.eth.limo

► MORE ON THE BACK: DEVELOPER URLS, EXAMPLES, NETWORK TOOLS, COMMUNITY ►


SWARM DEVELOPER RESOURCES
docs.ethswarm.org/docs/develop/resources

One curated landing page linking every category below. Bookmark it — links stay updated online, so this printed sheet keeps working.

LEARN
Get Started

your first 30 minutes

- What is Swarm [/concepts/what-is-swarm](#)
- Developer intro [/develop/introduction](#)
- Bee node quick-start [/bee/installation/quick-start](#)
- Swarm papers [papers.ethswarm.org](#)



SCAN → GET-STARTED HUB

Key Concepts

how it actually works

- DISC [/concepts/DISC](#)
- Incentives & BZZ [/concepts/incentives/overview](#)
- Postage stamps [/.../buy-a-stamp-batch](#)
- Feeds · Manifests · Chunk types



SCAN → DEEP-DIVE ON CONCEPTS

Developer Guides

recipes for common builds

- Upload & download [/develop/upload-and-download](#)
- Host a website [/develop/host-your-website](#)
- Routing · Dynamic content (feeds)
- Multi-author blog · Access control (ACT)



SCAN → ALL DEVELOPER GUIDES

EXAMPLES
Publishing & Websites

static sites, blogs, file uploads

- upload-react [ethersphere/examples-js](#)
- Simple Blog [ethersphere/examples](#)
- Basic Static Website [ethersphere/examples](#)



SCAN → MORE PUBLISHING EXAMPLES

Streaming

early HLS-over-Swarm work

- Stream React example [Solar-Punk-Ltd](#)
- Stream JS library [swarm-stream-js-lib](#)
- Reference: feeds [/develop/dynamic-content](#)



SCAN → STREAMING REPO

Messaging & Chat

PSS & GSOC patterns

- Chat React example [Solar-Punk-Ltd](#)
- Chat JS [swarm-chat-js](#)
- PSS · GSOC [/develop/.../pss](#) · [/gsoc](#)



SCAN → CHAT REPO

TOOLS
SDKs & APIs

talk to Bee from your code

- bee-js [bee-js.ethswarm.org](#)
- Bee HTTP API docs [docs.ethswarm.org/api](#)
- swarm-cli · Swarm MCP [ethersphere/swarm-mcp](#)



SCAN → BEE-JS DOCS

Gateways & Deploy

ship & inspect

- Bleeport app [ethswarm.org](#)
- Public gateway [gateway.ethswarm.org](#)
- Local test network · SwarmScan



SCAN → DEPLOY & GATEWAY

Community & Support

stuck? ask here first

- Discord [discord.ethswarm.org](#)
- Bee FAQ [/docs/bee/bee-faq](#)
- GitHub · Blog [ethersphere](#) · [blog.ethswarm.org](#)



SCAN → JOIN DISCORD